Isa Mukaramov

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SUMMARY

Mechanical Engineering graduate transitioning into software engineering, with experience converting ambiguous customer requirements into structured, tested and practical technical solutions. Brings a systems-oriented approach to problem solving, including requirements decomposition, iterative development, failure analysis, data-driven validation and technical documentation. Developing applications using Python and TypeScript and seeking to apply an engineering mindset within Westpac's Uncommon Minds Graduate Program.

EDUCATION

Western Sydney University

Bachelor of Mechanical Engineering, Relevant coursework: Control Systems, Engineering Design and Innovation, Engineering Mechanics, Mechanics of Materials, Materials Science, CAD and Technical Drawing — *Expected July 2026*

EXPERIENCE

Junior Mechanical Designer — Internship — February 2026 – Present

Troax

- Translate customer requirements into structured industrial fencing layouts, quotations and materials lists using CET Material Handling and the Troax Extension.
- Break down site, equipment and safety constraints into practical design configurations for customer and internal review.
- Coordinate with customers, production teams and manufacturing facilities to clarify requirements, identify conflicts and resolve implementation constraints.
- Review custom equipment requirements for feasibility, balancing customer expectations against manufacturing, installation and commercial limitations.
- Maintain accurate quotation, shipping and handover documentation, using Incoterms to define responsibilities, delivery processes and commercial expectations.
- Communicate technical information across customer, design and production stakeholders to support reliable project delivery.

Retail Sales Associate — October 2021 – November 2024

Supercheap Auto

- Diagnosed customer needs by asking structured questions about vehicle, product and usage requirements before recommending appropriate parts, tools and alternatives.
- Translated technical automotive information into clear guidance for customers with varying levels of technical knowledge.
- Compared products against compatibility, performance, cost and availability constraints to help customers make practical purchasing decisions.
- Developed strong customer communication and problem-solving skills in a fast-paced, high-volume environment.

PROJECTS

Handlebar Mount | System Design, Iterative Development, Load Testing and Failure Analysis

- Translated customer and functional requirements into a proprietary mounting system, progressing from initial concept to a validated design through ten structured development iterations.
- Developed and compared prototype variations using CAD, FDM printing, practical load testing and FEA feedback, using observed failure behaviour to identify high-stress regions and guide subsequent design changes.

- Established measurable acceptance criteria and validated the final design against a 400N load requirement.
- Evaluated ASA, ABS, 72D TPU and PA6-CF against factors including deformation resistance, layer adhesion and operating durability, selecting PA6-CF based on the project's performance constraints.
- Documented design decisions and testing outcomes across iterations, creating traceability between requirements, observed results and final implementation choices.

Intake Piping System | 3D Data Processing, Constraint Modelling and Fitment Validation

- Designed a scan-referenced intake system by converting physical engine-bay geometry into a workable digital model using 3D scanning, mesh processing and CAD.
- Processed and assessed multiple scan captures, selecting the most reliable dataset to reduce downstream modelling error and improve confidence in design decisions.
- Modelled routing around real packaging, clearance and component constraints rather than relying on assumed measurements.
- Used staged FDM prototypes to validate fitment incrementally, reducing the risk and cost of committing directly to the final design.
- Achieved interference-free vehicle fitment and measured a 1kPa static-pressure increase resulting from a 10% reduction in cross-sectional area, demonstrating the relationship between design choices and system performance.

Dolphin Torch Housing Redesign | Reverse Engineering, Interface Definition and Technical Documentation

- Reverse engineered an existing mechanical assembly to understand its components, interfaces, dimensional relationships and functional requirements.
- Reconstructed and redesigned the housing in SolidWorks, focusing on component alignment, fit, assembly behaviour and functional accuracy.
- Produced technical drawings with defined dimensions and tolerances to communicate implementation requirements clearly and support repeatable manufacturing.
- Applied tolerance standards across interacting parts to manage variation and reduce the risk of assembly or fitment failure.

TECHNICAL SKILLS

Programming: Python, TypeScript, MATLAB

Engineering Methods: requirements decomposition, system modelling, iterative development, testing and validation, root-cause analysis, technical documentation

Engineering and Design Tools: SolidWorks, CET Material Handling, 3D scanning, mesh processing, FDM prototyping, FEA interpretation

Commercial and Delivery: stakeholder communication, customer requirements, quotations, production coordination, materials lists, Incoterms, WHS